

Aditeya Baral

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EDUCATION

New York University, Courant Institute of Mathematical Sciences

Masters in Computer Science; GPA - 3.87/4.00

Concentration: Artificial Intelligence

New York City, USA

Sep 2024 – May 2026

PES University

Bachelor of Technology in Computer Science & Engineering; GPA - 8.71/10.00

Specialization: Machine Intelligence & Data Science

Bengaluru, India

Aug 2018 – May 2022

RESEARCH EXPERIENCE

Redis

Applied Research Scientist Intern, Redis LangCache; Advisor: Srijith Rajamohan

San Francisco, USA

June 2025 – Dec 2025

- Architected a *two-stage retrieval and re-ranking* pipeline for **Redis LangCache**, achieving a **12.5% PR-AUC** and **8% P-CHR AUC** improvement over baselines by integrating *cross-encoder re-rankers* for *full token-level interaction*.
- Curated and open-sourced **LangCache SentencePairs (v1-v3)**, a large-scale dataset family spanning **1M to 40M examples** from diverse linguistic sources, enabling robust fine-tuning of semantic retrieval and re-ranking models.
- Open-sourced **LangCache ReRanker v1** and **v2** model families comprising cross-encoder variants fine-tuned with ranking and classification objectives, enabling *application-specific score calibration* for diverse semantic caching use cases.
- Assisted in the fine-tuning and deployment of **LangCache Embed v3**, a generalist model for semantic retrieval, achieving **13.5% PR-AUC** improvement over v2 and outperforming larger general-purpose models even without re-ranking.
- Developed a *comprehensive evaluation framework* integrated with **RedisVL** for LangCache customers, enabling systematic analysis of achievable P-CHR tradeoffs, valid cache-hit rates, and operational thresholds before onboarding.
- Quantified *retriever bottlenecks* and *aggressive vs. conservative re-ranking effectiveness* by analyzing recall ceilings and re-ranking movement to *optimize operational trade-offs* and *improve cache-hit quality*.
- Supported *downstream integration* and development of **LMCache** by *building prototypes* and *conducting performance studies* with **Redis** as an *in-memory KV store*, demonstrating latency and throughput gains.

Computational Intelligence, Vision, and Robotics (CILVR) Lab

Research Assistant; Advisors: Shauli Ravfogel, Tal Linzen

New York City, USA

May 2025 – Present

- Investigating *arithmetic circuit dynamics* in LLMs when operators are redefined in-context by analyzing *activation representations* and *attention patterns* across transformer layers using **Llama-3.3-70B-Instruct**.
- Conducting *layer-wise analysis* of activation geometry using **PCA**, *centroid trajectories*, and *cluster separability metrics* to trace representational evolution under operator semantic redefinition.
- Examining *attention circuit reconfiguration* at token and head levels to determine whether semantic remapping reuses existing circuits or activates distinct computational pathways.

Computation and Psycholinguistics Lab

Research Assistant; Advisors: Jackson Petty, Tal Linzen

New York City, USA

May 2025 – Present

- Evaluating LLMs on *compositional generalization* and *instruction synthesis* by studying their ability to translate synthetic *Context-Free Grammars (CFGs)* into conforming strings.
- Analyzing model outputs in *few- and zero-shot settings* to assess *grammatical conformity* and uncover *generation strategies* used during translation.

Cisco Systems

Applied AI Engineer, Webex Media Quality Analytics

Bengaluru, India

July 2022 – July 2024

- Instruction fine-tuned LLMs like **Mistral** and **Llama-2** on-prem to enable *secure* and *cost-effective* AI solutions such as *translation* and *RAG* for engineers and customers, *cutting third-party dependency costs by 30%*.
- Led the initiative to build a novel *pre-training algorithm* for conversational data using **PyTorch** and **HuggingFace**, achieving a **40% performance gain** over standard approaches at benchmark fine-tuning tasks.
- Developed the **Webex Contextual Search** engine and *improved searching, ranking, recommendations* and *topic modelling* by **75%** with **<10%** increased overhead latency.
- Integrated **OpenAI** APIs and on-prem LLMs with the **Webex AI Assistant** for **15M+** worldwide users to add *auto-replies*, *summarisation*, *querying* and *action-item extraction* to message threads and meeting transcripts.

Intel Corporation

Applied Research Scientist Intern, Intel VSG; Advisors: Anay Majee, Anbumani Subramanian

Bengaluru, India

Aug 2021 – Dec 2021

- Explored **Few-Shot Learning Object Detection (FSOD)** techniques to reduce *catastrophic forgetting* in constrained and heterogeneous driving environments.

- Investigated and designed novel *representation learning* and *attention mechanisms* to learn *inter/intra-object relationships* using **PyTorch**.
- Outperformed existing approaches at the time on base and novel classes by **0.2 mAP** and **3 mAP** on the *Few-Shot India Driving Dataset*, a benchmark for FSOD.

Center for Cloud Computing and Big Data

Research Assistant; Advisor: KV Subramaniam

Bengaluru, India

May 2020 – July 2020

- Compiled and used **TailBench** to *simulate and profile* application loads, monitor performance, and analyse results.
- Explored ways to *reduce tail latencies* in latency-critical applications such as translation and image recognition.

SOFTWARE ENGINEERING EXPERIENCE

Cisco Systems

Big Data Engineer, Webex Media Quality Analytics

Bengaluru, India

July 2022 – July 2024

- Developed and deployed streaming jobs in **Scala** and **Flink** to process **1M+ reports/min** and compute **1200+ real-time metrics** from Calls and Meetings.
- Applied *statistical modelling* techniques to investigate and report *media quality insights* to downstream consumers, *reducing errors by 30%* and *analysis time by 15 hrs/week* per team member.
- Led the development of *real-time (<1 min) auditing pipelines* using **Kafka** and **Python** to ensure *per-minute data consistency* between streaming jobs and **Iceberg** and **Pinot** data stores, *reducing manual effort by >80%*.
- Built graphs and dashboards on the **Webex Media Quality Analytics Dashboard** using **Grafana** and **Kibana** to set up alerts and KPIs for **20,000+** clients and customers.

Big Data Engineering Intern, Webex VideoMesh Analytics

Jan 2022 – June 2022

- Migrated the **Meetings Analytics Engine** from Java and Spark to **Scala** and **Flink** to scale up to **1M+ reports/min** and significantly *improve real-time report generation* by over **40%**.
- Built **VideoMesh Developer APIs** using **Java** and globally rolled them out for **30,000+ enterprises** with **customer-facing applications**.

PREPRINTS AND PROJECTS

- [1] **When ‘+’ Means ‘-’? Probing Arithmetic Circuits Under Symbol Redefinition**
Authors: Aditeya Baral, Allen George Ajith, Shauli Ravfogel
- [2] **Can LLMs *understand* Math? Exploring the Pitfalls in Mathematical Reasoning**
Authors: Tiasa Singha Roy*, Aditeya Baral*, Ayush Rajesh Jhaveri, Yusuf Baig
- [3] **CMLFormer – A Dual Decoder Transformer with Switching Point Learning for Code-Mixed Language Modeling**
Authors: Aditeya Baral, Allen George Ajith, Roshan Nayak, Mrityunjay Abhijeet Bhanja
- [4] **Patch and Control – Steering Behavior of Large Vision-Language Models via Latent Activations**
Authors: Aditeya Baral, Rijul Dahiya, Dilip Venkatesh

PAPERS AND PUBLICATIONS

- [1] **Training for Compositional Sensitivity Reduces Dense Retrieval Generalization**
ICLR 2026, Sci4DL
Authors: Radoslav Ralev, Aditeya Baral, Iliya Zhechev, Jen Agarwal, Srijith Rajamohan
- [2] **ChatBERT – Multi-task approach to Pre-Training for Structured Conversations**
Webex AI 2023
Authors: Aditeya Baral (Work done as part of Cisco Webex AI Research)
- [3] **CalBERT – Code-mixed Adaptive Language Representations using BERT**
AAAI 2022, MAKE
Authors: Aditeya Baral, Aronya Baksy, Ansh Sarkar, Deeksha D, Ashwini M Joshi
- [4] **Information Maximization to Overcome Catastrophic Forgetting in Few-Shot Object Detection**
Intel VSG Research 2021
Authors: Aditeya Baral, Anay Majee, Anbumani Subramanian

[5] **MAPLE – MAsking words to generate blackout Poetry using Seq2Seq LLearning**

ACL 2021, ICNLSP

Authors: Aditeya Baral, Himanshu Jain, Deeksha D, Mamatha H R

[6] **Analysis of Kepler Objects of Interest using ML for Exoplanet Identification**

IEEE CONIT 2021

Authors: Ameya Rajendra Bhamare, Aditeya Baral, Saarthak Agarwal

TEACHING EXPERIENCE

Teaching Assistant, CS322: Big Data

Bengaluru, India

PES University; Professors: Dr. KV Subramaniam, Dr. Prafullata K Auradkar, Animesh Giri

June 2021 – Dec 2021

Designed and graded coursework, assignments and projects, and delivered hands-on sessions on Hadoop and Spark for a class of 600+ enrolled students for the undergraduate Big Data course.

AWARDS AND SCHOLARSHIPS

Webex Analytics Datathon, Cisco Systems

2nd / 20+ teams

2024

Webex IDEA Hackathon, Cisco Systems

1st / 300+ teams

2023

Webex Playtime Hackathon, Cisco Systems

Top 20 Globally & Regional Winner / 300+ teams

2023

Undergraduate Researcher Award, PES University

One among 900+ students

2022

Prof. CNR Rao, MRD & DAC Scholarship Awards

Top 20% among 900+ students

2022

Finalist, Intel Technovation, Flipkart, IBM, IISc Hackathons

One among 200+ teams

2022

National newspaper coverage, ToI

Remodelled garbage collection tracking in BLR

2017

SKILLS

Languages: Python, Scala, Java, C/C++, R, Groovy, SQL, L^AT_EX

ML/Stats Libraries: PyTorch, Tensorflow, HuggingFace, NLTK, pandas, NumPy, scikit-learn, seaborn, matplotlib, plotly

AI/ML Techniques: Representation Learning, Mechanistic Interpretability, Transfer Learning, Language Models, RAG

Big Data/Cloud: Hadoop, Kafka, Zookeeper, Spark, Flink, Iceberg, Pinot, Redis, ELK

Frameworks/Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Flask, Grafana, PSQl, MongoDB, AWS, Linux

SERVICES AND VOLUNTEERING EXPERIENCE

Speaker, Guest Lecture on - Building Foundation Models using Transformers

Bengaluru, India

PES University

Sep 2023

Delivered a guest lecture to undergraduate students on the advancements in representation learning techniques for language and highlighted the importance of interdisciplinary research.